Shubhendra Pratap Singh — Curriculum Vitae

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Professional Summary

Mechanical Engineer and product innovator with 9+ *years of combined industry and academic research* spanning rapid prototyping, mechanical system design, test/validation, and automation. Current graduate research centers on *AI-powered critical-chain modeling for bio-inspired analogy matching*—a function–flow–performance (FFP) framework with a prototype LLM replication for reliability checks—aimed at democratizing bio-inspired design and turning biology-inspired ideas into adoptable engineering solutions. *Portfolio includes*: emissions lab automation (≈approximately $600k annual savings), ECU IP benchmarking studies, rural water purification, service robotics, Li-ion diagnostics, and humane wildlife tagging devices. Experienced bridging concept ↔ production with *functional modeling, rapid prototyping, and evidence-backed pipelines*.

Education

Clemson University — *M.S., Mechanical Engineering* | Aug 2023 – Aug 2025 (Expected)
GPA: 3.85 · IDEAS^3 Lab · Advisor: Dr. Cameron Turner
Thesis: *Identifying Adopted Strategies in Bio-Inspired Design Through Critical Chain and Function–Flow–Performance Matching with an AI-Augmented Framework* (target Aug 2025).

- Built a two-part framework: (1) rule-guided critical-chain extraction from function structures, and (2) FFP analogy matching.
- Integrated a prototype LLM "like-for-like" replication to measure reliability (overlap, edit distance, pruning concordance).
- Long-term goal: expand framework across electromechanical, marine, and cybersecurity systems.

JECRC University — *B.Tech, Mechanical Engineering* | Aug 2012 – Aug 2016
Founder, *JU Automotive Club* (grew 5 → 70 members); led teams at national design competitions (Eco-Kart, Effi-Cycle, Quad Bike, Electric Baja, Formula Student). Awarded *Best Design* (e-kart).

Professional Experience

Graduate Research Assistant (GRA) — Clemson University | Aug 2025 – Dec 2025
Project 1: Enhancing Track Vehicle Design Engineering Through Digital Twins (Army GVSC collaboration)

- Worked on evaluating non-functional requirements (-ilities) for tracked vehicle systems.

- Identified and provided relevant -ilities (e.g., Maneuverability, Trafficability, Climb and Descend Slopes, Transportability, Laterally Traverse Slopes) with formal definitions and mathematical expressions.

Project 2: Case Study in Graduate Vehicle Design: Effects of Targeted Instruction on Non-Functional Requirement Understanding

- Qualitative analyses of semi-structured interviews and focus groups; co-authoring manuscript.
- Completed IRB Group 1 SBR training.

Graduate Grading Assistant / Teaching Assistant (TA) — Clemson University | Aug 2024 – May 2025

- Courses: ME 6720 (Manufacturing Optimization), ME 8700 (Advanced Design Methodology), ME 4010 (Engineering Design), ME 3060 (Fundamentals of Machine Design).

Project Engineer — FEV India (Pune, India) | Nov 2017 – Apr 2021

- Piloted/chassis-dyno testing & calibration across Euro-6 diesel, SCR emission systems, and hybrid engines (Cummins, John Deere, Mahindra).
- Developed an automation tool combining three manual processes; reduced test-pipeline runtime from 30 min → 2–3 min, saving ≈$600k annually.
- Tools: ETAS INCA, Cummins Calterm, MATLAB/Simulink, AVL FTIR, Horiba MEXA.

Research Associate — Club First (Robotics & Automation), Jaipur | Jul 2016 – Nov 2017

- Designed & validated Instant Water Purification & Vending System (IWPVS).
- Built Serving Robot mechanical structure + dynamics/kinematics (validated via FEA).

Engineering Intern (AE) — Honda Cars India Ltd., Greater Noida | Jan 2016 – Jun 2016

- Designed a “Master Jig” eliminating petrol/diesel changeover (38.5s loss → 0s).
- Reduced operator count and downstream interferences; advanced to supplier prototype.

Publications & Presentations

SAE WCX 2026 (Accepted Abstracts)

- Sutton, M.; Anbuvanan, A.; Singh, S. P.; Castanier, M. P.; Turner, C.; Kurz, M. E. *Effects of Targeted Instruction on Non-Functional Requirement Understanding*.
- Suber, D.; Bradley, A.; Singh, S. P.; Turner, C.; Wagner, J.; Castanier, M. P. *Enhancing Track Vehicle Design Engineering Through Digital Twins*.

ASME IDETC-CIE 2025 (Accepted Poster Presentation)

- Singh, S. P.; Turner, C. *Identifying Adopted Strategies in Bio-Inspired Design Through Critical Chain and Function–Flow–Performance Matching.*
  Poster accepted at ASME-CIE Graduate Research Poster Session, IDETC-CIE 2025.

Journal Manuscript (In Preparation)

- Singh, S. P.; Turner, C. *Identifying Adopted Strategies in Bio-Inspired Design Through Critical Chain and Function–Flow–Performance Matching.* Target: *ASME Journal of Mechanical Design*.

Other Projects & Research

- **Whale-Bot Tagging Device** (2024–2025): Bio-inspired, non-invasive marine tracking tag. Harnesses whale's kinetic energy for self-powered long-duration tagging (patent filing in process).
- **ECU IP-Theft Study** *(Georgia Tech collab)* (2023-2024): Demonstrated risks of ECU benchmarking; tied to design IP theft concerns. (Presented to Dr. Saman Zonouz and Dr. Animesh Chhotray)
- **Emissions-Cycle Automation** *(FEV)* (2017–2021): Macro-based tool, cut runtime and reporting, still in service.
- **Vitrification Automation** *(CEDAR Lab & IDEAS^3 Lab)* (2023–2024): SolidWorks + 3D printing for tissue cassette automation.
- **Li-ion Battery OBD** (Clemson, 2023–2024): EKF-based diagnostics and hybrid thermal-electrical modeling.
- **IWPVS Rural Water System** (Club First, 2016–2017).
- **Service Robot for Healthcare Logistics** (2016).

Relevant Coursework

Advanced Design Methodology · Engineering Optimization · Statistics with R · Data Visualization · On-Board Diagnostics · Material Selection · Design of Automation · Modeling & Simulation · Design for Manufacturing · Finite Element Formulation

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Technical Skills

- **CAD:** SolidWorks, Fusion 360, Inventor, ANSYS
- **Rapid Prototyping:** 3D printing (FDM/SLA), laser cutting, jig/fixture design
- **Frameworks & Tools:** DFMEA, functional-basis modeling, tolerance analysis
- **Programming & Simulation:** MATLAB, Simulink, Python, C++, R
- **Testing/Diagnostics:** Engine dynos, climate chambers, emissions analyzers

- **Systems:** ETAS INCA, Cummins Calterm, Horiba MEXA, SCADA, PLC

Affiliations, Leadership & Outreach

- **Professional Memberships:** Member of SAE International.
- **Research Affiliations:**
  - **CEDAR Lab** (Clemson Engineering Design Application and Research Lab).
  - **Deep Orange**, Clemson University's accelerated vehicle concept development program (Dept. of Automotive Engineering).
  - **VIPR-GS** *(Virtual Prototyping of Autonomy-Enabled Ground Systems)* – research effort on autonomy-enabled mobility and digital prototyping for Army platforms.
- **Outreach & Teaching:** Served as Guest Lecturer/Resource Person on Rapid Prototyping and Robotics at Regional College of Education, Research and Technology (2-month tenure) and conducted multi-week workshops at leading institutions across India, including Banasthali Vidyapith, JECRC University, and several others.
- **Creative/Photography:** Member of the *Clemson Photography Club*; images featured on Clemson University's official Instagram page.
- **University Consortium:** Member of the *Tigers United University Consortium;* led the official collaboration between Clemson Engineering and Tigers United to integrate conservation-focused technology (e.g., whale-tagging concept development).